



INSTA MECHANIC – ONLINE VEHICLE REPAIR & ASSISTANCE PLATFORM

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CERTIFICATE

This is to certify that the project synopsis entitled “Insta Mechanic – Online Vehicle Repair & Assistance Platform” is submitted by Kumar Raj Ranjan, Vihan Mudgal, and Umesh Roy as the minor project of VI th Semester Computer Science & Engineering is a bonafide work carried out under the guidance and supervision of the concerned faculty.

The work presented in this synopsis is original and has not been submitted elsewhere for the award of any other degree or diploma.

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ABSTRACT

Insta Mechanic is an innovative web-based vehicle repair and roadside assistance platform developed to simplify the process of connecting vehicle owners with nearby verified mechanics in real time. The platform is designed to provide quick and reliable assistance during vehicle breakdowns, emergencies, and routine maintenance requirements.

The proposed system provides separate dashboards for customers, mechanics, and administrators. Customers can search and book mechanics, track booking status, and make online payments. Mechanics can update availability, accept or reject service requests, and manage service progress. Administrators can verify mechanics, monitor platform activities, and maintain transparency across the system.

The platform is developed using modern web technologies including HTML5, CSS3, Bootstrap 5, JavaScript, Node.js, Express.js, PostgreSQL, Socket.io, JWT Authentication, and online payment integration systems such as Razorpay and Stripe.

The project aims to modernize the traditional vehicle servicing process by introducing real-time communication, transparency, digital payments, and automated service management.

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1. INTRODUCTION

The automobile service industry has witnessed significant growth over the past decade. However, many vehicle repair and roadside assistance services still rely on traditional communication methods such as phone calls and local referrals. This creates inconvenience for customers during emergencies and often leads to delays in receiving assistance.

Insta Mechanic is developed as a digital solution to bridge the gap between customers and mechanics through a centralized online platform. The system enables customers to locate nearby verified mechanics, book services, track repair progress, and complete secure online transactions.

The project aims to improve efficiency, accessibility, transparency, and reliability in vehicle servicing through the use of modern web technologies and real-time communication systems.

2. BACKGROUND STUDY

Traditional vehicle repair systems are highly dependent on local garages and physical communication. During emergencies, users often face difficulties in locating trusted mechanics quickly. Existing systems generally lack live tracking, digital booking management, payment integration, and customer support mechanisms.

With the rise of online service platforms and mobile-based solutions, there is a growing demand for digital automobile service management systems. Applications such as ride-booking and food delivery platforms have inspired similar on-demand service systems for the automobile industry.

Insta Mechanic is inspired by these modern service models and aims to provide an efficient and user-friendly platform specifically focused on vehicle repair and roadside assistance.

3. PROBLEM STATEMENT

Vehicle owners face several challenges while searching for repair and maintenance services:

- Difficulty in locating nearby mechanics during emergencies.
- Lack of trust and transparency in service providers.
- Delayed response times.
- No centralized service booking platform.
- Lack of live service tracking.
- No secure online payment system.
- Inefficient communication between customers and mechanics.

The proposed system aims to solve these issues by developing a centralized digital platform for vehicle repair and roadside assistance.

4. OBJECTIVES OF THE PROJECT

The primary objectives of the project are:

- To develop an online platform for vehicle repair and roadside assistance.
- To connect users with nearby verified mechanics.
- To provide real-time booking and tracking facilities.
- To integrate secure online payment gateways.
- To create role-based dashboards for customers, mechanics, and administrators.
- To improve service transparency and communication.

- To reduce response time during emergencies.
 - To maintain digital service records and booking history.
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5. SCOPE OF THE PROJECT

The scope of Insta Mechanic includes:

- Online mechanic booking system.
- Emergency roadside assistance.
- Real-time service tracking.
- Online payment and invoice generation.
- Mechanic availability management.
- Admin monitoring and verification.
- Customer feedback and review system.

The system can initially be deployed within a single city and later expanded to multiple regions.

6. EXISTING SYSTEM

The current system mainly depends on traditional garage services and manual communication methods. Customers usually search for mechanics through personal references or nearby shops.

Limitations of Existing System

- No digital booking management.

- Lack of real-time tracking.
 - No proper verification of mechanics.
 - Time-consuming process.
 - Lack of transparency in pricing.
 - No centralized database for customer records.
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7. PROPOSED SYSTEM

The proposed system is a responsive web application that provides a digital platform for customers and mechanics.

Features of Proposed System

- User Registration and Login.
- Nearby Mechanic Search.
- Booking and Scheduling.
- Live Status Tracking.
- Secure Payment Gateway.
- Notifications and Alerts.
- Mechanic Dashboard.
- Admin Panel.
- Booking History.
- Ratings and Reviews.

The platform will be mobile-friendly and accessible across different devices.

8. FEASIBILITY STUDY

Technical Feasibility

The project is technically feasible because the required technologies such as Node.js, PostgreSQL, Bootstrap, and JavaScript are open-source and widely available.

Economic Feasibility

The development cost is minimal because most technologies and hosting services used are affordable or free for educational purposes.

Operational Feasibility

The system is user-friendly and designed to simplify service booking and communication.

9. SYSTEM REQUIREMENTS

Hardware Requirements

- Intel i3 Processor or Higher
- Minimum 4GB RAM
- 20GB Storage Space
- Stable Internet Connection

Software Requirements

- Windows/Linux/macOS
- Visual Studio Code
- Node.js Runtime Environment
- PostgreSQL Database
- Web Browser
- Postman API Testing Tool

10. TECHNOLOGIES USED

Frontend Technologies

- HTML5
- CSS3
- Bootstrap 5
- JavaScript

Backend Technologies

- Node.js
- Express.js

Database

- PostgreSQL

Additional Technologies

- Socket.io
 - JWT Authentication
 - REST APIs
 - Razorpay / Stripe Payment Gateway
 - Prisma ORM / Sequelize
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11. SYSTEM ARCHITECTURE

The system architecture follows a client-server model.

Components

Frontend Layer

Handles user interaction and interface.

Backend Layer

Processes requests, manages APIs, authentication, and business logic.

Database Layer

Stores user data, bookings, payments, and mechanic information.

Payment Gateway

Handles secure online transactions.

Real-Time Communication Layer

Socket.io is used for real-time booking and status updates.

12. MODULES OF THE SYSTEM

Customer Module

Features

- Registration and Login
- Search Nearby Mechanics
- Book Services
- Make Payments
- Track Booking Status
- View Booking History

Mechanic Module

Features

- Profile Management
- Availability Management
- Accept/Reject Requests
- Update Service Status
- Earnings Dashboard

Admin Module

Features

- Verify Mechanics
 - Manage Users
 - Monitor Bookings
 - Generate Reports
 - Manage Complaints
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13. DATABASE DESIGN

The database stores all information related to users, mechanics, bookings, and payments.

Main Tables

Users Table

- User ID
- Name
- Email
- Password
- Phone Number
- Role

Mechanics Table

- Mechanic ID
- Availability Status
- Experience
- Verification Status

Bookings Table

- Booking ID
- Customer ID
- Mechanic ID
- Service Type
- Booking Status
- Payment Status

Payments Table

- Payment ID
 - Booking ID
 - Amount
 - Payment Method
 - Transaction Status
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14. METHODOLOGY

The project follows Agile Software Development Methodology.

Phases

Phase 1 – Requirement Analysis

Understanding system requirements and user expectations.

Phase 2 – Design

Creating wireframes, database design, and architecture.

Phase 3 – Development

Frontend and backend implementation.

Phase 4 – Testing

Unit testing, API testing, and integration testing.

Phase 5 – Deployment

Hosting the application on cloud platforms.

Phase 6 – Maintenance

Bug fixing and future improvements.

15. FUNCTIONAL REQUIREMENTS

- User Registration and Login.
 - Booking Management.
 - Mechanic Availability Management.
 - Payment Processing.
 - Real-Time Notifications.
 - Admin Monitoring.
 - Service Tracking.
 - Customer Reviews and Ratings.
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16. NON-FUNCTIONAL REQUIREMENTS

Performance

The system should provide quick response times.

Security

User data and payments must be secure.

Scalability

The system should support future expansion.

Reliability

The platform should operate continuously with minimal downtime.

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17. TESTING STRATEGY

Unit Testing

Testing individual modules.

Integration Testing

Testing interaction between modules.

API Testing

Testing backend APIs using Postman.

User Acceptance Testing

Ensuring the platform meets user expectations.

18. SECURITY FEATURES

- Password Hashing using bcrypt.
- JWT Authentication.
- Role-Based Access Control.

- Input Validation.
 - Secure Payment Gateway Integration.
 - HTTPS Communication.
 - Rate Limiting and API Security.
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19. ADVANTAGES OF THE PROPOSED SYSTEM

- Fast and reliable service booking.
 - Improved customer convenience.
 - Real-time tracking and communication.
 - Secure online payment integration.
 - Better transparency in services.
 - Reduced manual effort.
 - Efficient booking management.
 - Increased trust through mechanic verification.
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20. LIMITATIONS

- Initial deployment limited to specific regions.
 - Dependence on internet connectivity.
 - Real-time tracking may require additional GPS support.
 - Payment gateway dependency.
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21. FUTURE ENHANCEMENTS

- Android and iOS Mobile Applications.
 - AI-Based Service Recommendations.
 - GPS Navigation Integration.
 - Voice Assistant Support.
 - Multi-language Support.
 - Subscription and Membership Plans.
 - Analytics Dashboard.
 - Automatic Emergency Detection.
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22. CONCLUSION

Insta Mechanic is an innovative and scalable solution designed to modernize the vehicle repair and roadside assistance industry. The system integrates online booking, secure payments, real-time communication, and mechanic management into a single platform.

The proposed solution has the potential to improve accessibility, transparency, and efficiency in vehicle servicing while providing a convenient experience for customers and mechanics alike.

By leveraging modern web technologies and scalable architecture, the project can evolve into a comprehensive nationwide automobile service platform in the future.

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